

Transfer Learning Through Conditional Quantile Matching

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Objective: Predict the country-level MAU of a target app using data from Meta's family of apps.

Regression Task: Predict the country-level MAU $Y^{(0)} \in \mathcal{Y}$ for a target app from country-specific covariates $X^{(0)} \in \mathcal{X}$.

- 1 **Source-Domain Data:** $\mathcal{D}_S^{(k)} = \left\{ \left(X_i^{(k)}, Y_i^{(k)} \right) \right\}_{i=1}^{n_k} \sim P^{(k)}$ for $k = 1, \dots, K$.
- 2 **Target-Domain Data:** $\mathcal{D}_T = \left\{ \left(X_i^{(0)}, Y_i^{(0)} \right) \right\}_{i=1}^{n_0} \sim P^{(0)}$, where $n_0 \ll n_k$ for $k = 1, \dots, K$.

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Covariate Shift (Shimodaira, 2000):

$$P^{(k)}(Y|X) = P^{(0)}(Y|X) \text{ but } P^{(k)}(X) \neq P^{(0)}(X).$$

Label Shift (Saerens et al., 2002):

$$P^{(k)}(X|Y) = P^{(0)}(X|Y) \text{ but } P^{(k)}(Y) \neq P^{(0)}(Y).$$

General Target Shift:

$$P^{(k)}(X) \neq P^{(0)}(X) \quad \text{and} \quad P^{(k)}(Y|X) \neq P^{(0)}(Y|X) \quad \text{for } k = 1, \dots, K.$$

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1 Learn source distributions

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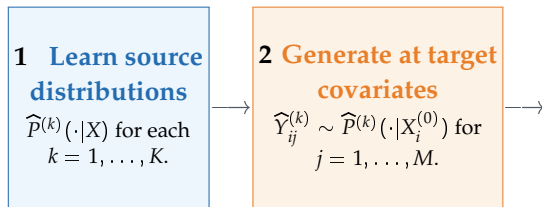


- ① Learn a generative model $\hat{P}^{(k)}(\cdot|x)$ using $\left\{ \left(X_i^{(k)}, Y_i^{(k)} \right) \right\}_{i=1}^{n_k}$ for each source domain k .
 - Model $P^{(k)}(\cdot|x) = g^{(k)}(x, \cdot)_{\#} P_{\eta}$ and estimate $\hat{g}^{(k)}$ via engrression (Shen and Meinshausen, 2025):

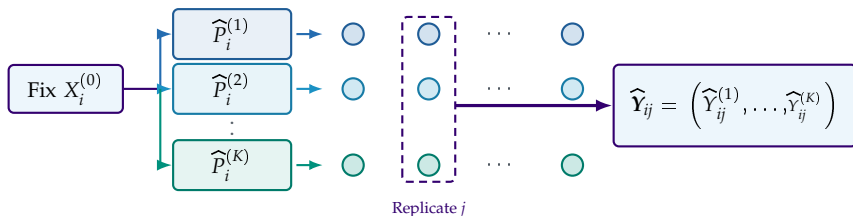
$$\hat{g}^{(k)} \in \arg \min_{g \in \mathcal{G}} \frac{1}{n_k} \sum_{i=1}^{n_k} \left[\frac{1}{m} \sum_{j=1}^m \left| Y_i^{(k)} - g(X_i^{(k)}, \eta_{ij}) \right| - \frac{1}{2m(m-1)} \sum_{j=1}^m \sum_{j'=1}^m \left| g(X_i^{(k)}, \eta_{ij}) - g(X_i^{(k)}, \eta_{i,j'}) \right| \right],$$

where P_{η} is a prespecified noise distribution and $\mathcal{G}$ is a neural network class.

Proposed Transfer Learning Framework (TLCQM)

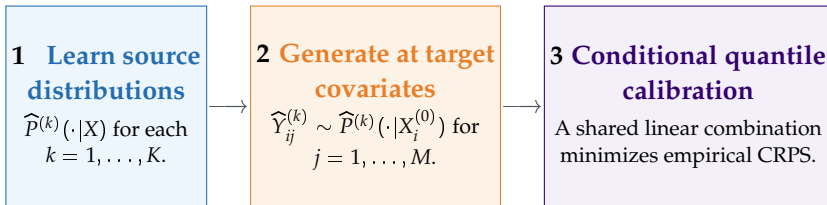


- 2 For each target covariate $X_i^{(0)}$, independently draw M synthetic response vectors:



$$\left\{ (X_i^{(0)}, \hat{Y}_{ij}) : i = 1, \dots, n_0, j = 1, \dots, M \right\}, \quad \hat{Y}_{ij} = (\hat{Y}_{ij}^{(1)}, \dots, \hat{Y}_{ij}^{(K)}), \quad \hat{Y}_{ij}^{(k)} \sim \hat{P}^{(k)}(\cdot|X_i^{(0)}).$$

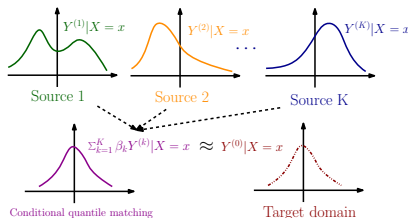
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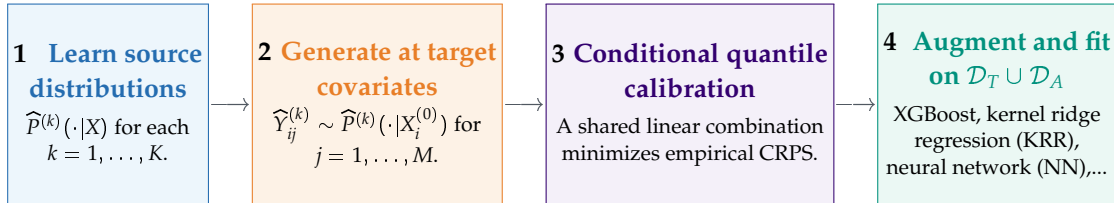
- 3 Compute the conditional quantile calibration estimator by minimizing the continuous ranked probability score (CRPS):

$$\hat{\beta} \in \arg \min_{\beta \in \Theta} \frac{1}{n_0} \sum_{i=1}^{n_0} \left\{ \frac{1}{M} \sum_{j=1}^M |Y_i^{(0)} - \beta^T \hat{V}_{ij}| - \frac{1}{2M(M-1)} \sum_{j,j'=1}^M |\beta^T \hat{V}_{ij} - \beta^T \hat{V}_{ij'}| \right\}.$$

- $\hat{V}_{ij} = (1, \hat{Y}_{ij})^T$, and $Y_i^{(0)}$ is matched with draws generated at the same target covariate $X_i^{(0)}$.



Proposed Transfer Learning Framework (TLCQM)



4 Augment the target domain with synthetic, target-calibrated labels:

$$\mathcal{D}_A = \bigcup_{k=1}^K \left\{ \left(X_i^{(k)}, \hat{\boldsymbol{\beta}}^T \hat{\mathbf{V}}_i^{(k)} \right) \right\}_{i=1}^{n_k} \quad \text{with} \quad \hat{\mathbf{V}}_i^{(k)} = \left(1, \hat{Y}_i^{(1,k)}, \dots, \hat{Y}_i^{(K,k)} \right)^T \in \mathbb{R}^{K+1},$$

where $\hat{Y}_i^{(j,k)}$ is a prediction of $Y^{(j)}$ at $X_i^{(k)}$, such as $\hat{\mathbb{E}}(Y^{(j)}|X_i^{(k)})$.

5 (Optional) Estimate $w_k(x) = \frac{dP_X^{(0)}(x)}{dP_X^{(k)}(x)}$ for $k = 1, \dots, K$ to correct for covariate shift.

Theory: Excess Risk Decomposition

- $R(f) = \mathbb{E}_{P^{(0)}}[\ell(Y^{(0)}, f(X^{(0)}))]$ and $f^{(0)} \in \arg \min_{f \in \mathcal{F}} R(f)$ under a loss function ℓ .
- $\text{Rad}_n(\mathcal{F})$ is the Rademacher complexity of $\mathcal{F}$.
- Define the conditional CRPS as $\hat{\Delta}(\hat{\boldsymbol{\beta}}) := \mathbb{E}_{X^{(0)}} \left\{ \int_{\mathbb{R}} \left[F^{(0)}(t|X^{(0)}) - \hat{G}_{\hat{\boldsymbol{\beta}}}(t|X^{(0)}) \right]^2 dt \right\}$.

Target-Only: $\hat{f}^{(0)} = \arg \min_{f \in \mathcal{F}} \frac{1}{n_0} \sum_{i=1}^{n_0} \ell(Y_i^{(0)}, f(X_i^{(0)}))$ satisfies

$$R(\hat{f}^{(0)}) - R(f^{(0)}) \lesssim \text{Rad}_{n_0}(\mathcal{F}) + \sqrt{\frac{\log(2/\delta)}{n_0}} \quad \text{w. p.} \geq 1 - \delta.$$

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TLCQM: $\hat{f}^{(0,tl)} = \arg \min_{f \in \mathcal{F}} \frac{1}{N} \left\{ \sum_{i=1}^{n_0} \ell(Y_i^{(0)}, f(X_i^{(0)})) + \sum_{k=1}^K \sum_{i=1}^{n_k} \hat{w}_k(X_i^{(k)}) \cdot \ell(\hat{\boldsymbol{\beta}}^T \hat{\mathbf{v}}_i^{(k)}, f(X_i^{(k)})) \right\}$ satisfies

$$R(\hat{f}^{(0,tl)}) - R(f^{(0)}) \lesssim \underbrace{\text{Rad}_N(\mathcal{F}) + \sqrt{\frac{\log(1/\delta)}{N}}}_{\text{Standard generalization error}} + \underbrace{\frac{1}{N} \sum_{k=1}^K \|\hat{w}_k - w_k\|_1}_{\text{Importance weight error}} + \underbrace{\frac{(N - n_0)^2 \cdot \hat{\Delta}(\hat{\boldsymbol{\beta}})}{N^2}}_{\text{Conditional transfer error}}$$

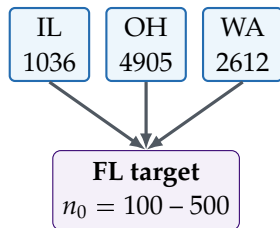
w. p. $\geq 1 - \delta$, where $N = \sum_{k=0}^K n_k$.

Let $\beta_* \in \arg \min_{\beta \in \Theta} \Delta(\beta)$, where $\Delta(\beta)$ is the population conditional CRPS discrepancy.

$$\begin{aligned} \hat{\Delta}(\hat{\beta}) \lesssim & \underbrace{\Delta(\beta_*)}_{\text{Transfer bias}} + \underbrace{\sum_{k=1}^K \|\hat{g}^{(k)} - g^{(k)}\|_{\infty}}_{\text{Source generator error}} + \underbrace{\sqrt{\frac{[\log n_0 + \log(1/\delta)] \left[\Delta(\beta_*) + \sum_{k=1}^K \|\hat{g}^{(k)} - g^{(k)}\|_{\infty} \right]}{n_0}}}_{\text{Localized target-sample error}} \\ & + \underbrace{\frac{\log n_0 + \log(1/\delta)}{n_0} + \epsilon_{\text{opt}} + O\left(\frac{1}{\sqrt{M}}\right)}_{\text{Optimization and Monte Carlo approximation errors}}. \end{aligned}$$

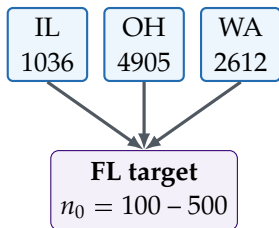
- ① $\Delta(\beta_*)$ is the **irreducible transfer bias**: the best conditional distribution approximation attained by the calibrated source family.
- ② It vanishes when $P^{(0)}(Y|X)$ lies in the **affine span** of the source conditional laws.
- ③ Large $\Delta(\beta_*)$ can outweigh variance reduction and cause **negative transfer**.

Goal: Predict apartment rental prices in FL using data from IL, OH, and WA.



500 Monte Carlo replications.

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MSE reduction versus target-only training

n_0	100	200	300	500
XGBoost	12.0%	5.5%	1.7%	-2.7%
KRR	78.5%	66.0%	57.9%	47.8%
NN	80.0%	65.9%	49.6%	34.1%

Result: MSE reductions are larger for more flexible learners (NN and KRR) and generally decrease as n_0 increases.



Objective: Predict country-level MAU for a held-out app at Meta.

- Four source apps with ≈ 230 observations per app across two device platforms.

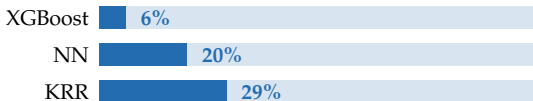


Objective: Predict country-level MAU for a held-out app at Meta.

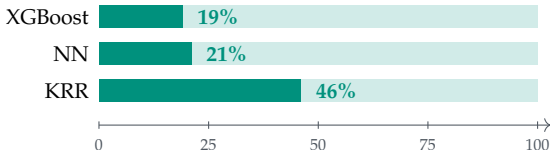
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MSE reduction from target-only to TLCQM

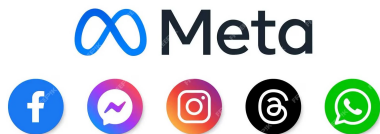
Platform I



Platform II



Monthly Active User Prediction at Meta

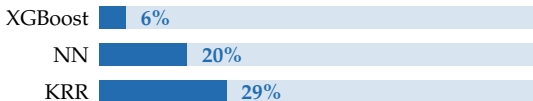


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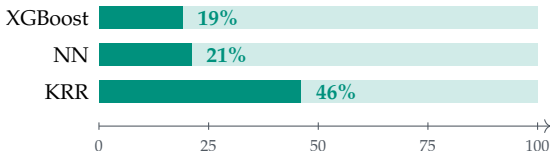
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MSE reduction from target-only to TLCQM

Platform I



Platform II



Average test MSE (standard errors in parentheses)

	XGBoost	TLCQM NN	KRR	TKRR	CDAR	DARC
Platform I	4.400 (0.500)	4.343 (0.481)	3.726 (0.440)	4.658 (0.482)	4.735 (0.527)	5.048 (0.532)
Platform II	2.886 (0.244)	2.812 (0.247)	2.036 (0.181)	2.937 (0.274)	3.467 (0.315)	3.884 (0.296)

Main Takeaways of TLCQM

- 1 **Generate conditional distributions**, from each source at target covariates.
- 2 **Calibrate with conditional CRPS** to align the source-generated and target response distributions.
- 3 **Train a standard predictor** on a target-aligned augmented dataset.

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Thank you!

More details are in

<https://arxiv.org/abs/2602.02358>.

Bottom Line: TLCQM improves target prediction when conditional transfer bias is small, but persistent bias can cause negative transfer.

- M. Saerens, P. Latinne, and C. Decaestecker. Adjusting the outputs of a classifier to new a priori probabilities: a simple procedure. *Neural Computation*, 14(1):21–41, 2002.
- N. Sgouropoulos, Q. Yao, and C. Yastremiz. Matching a distribution by matching quantiles estimation. *Journal of the American Statistical Association*, 110(510):742–759, 2015.
- X. Shen and N. Meinshausen. Engression: extrapolation through the lens of distributional regression. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 87(3):653–677, 2025.
- H. Shimodaira. Improving predictive inference under covariate shift by weighting the log-likelihood function. *Journal of Statistical Planning and Inference*, 90(2):227–244, 2000.
- Y. Zhang, S. Wilkins-Reeves, W. Lee, and A. Hofleitner. Transfer learning through conditional quantile matching. *arXiv preprint arXiv:2602.02358*, 2026.

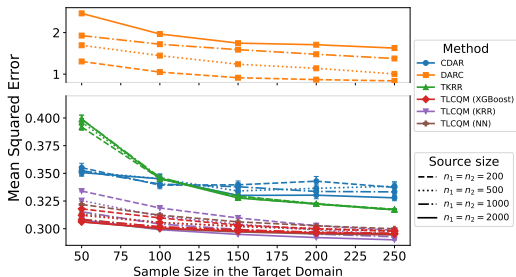
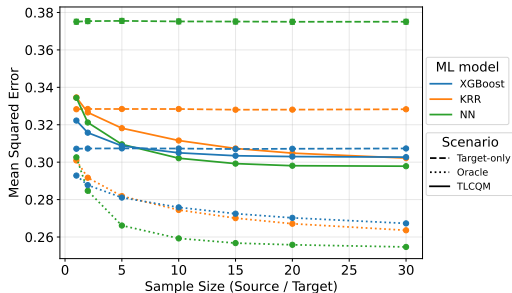
Simulation: General Target Shift

- Two Source Domains:

$$Y^{(1)} = \sin \left(3\theta^T X^{(1)} \right) + 1 + \epsilon, \quad Y^{(2)} = 2 \cos \left(3\theta^T X^{(2)} \right) + 1 + \epsilon,$$

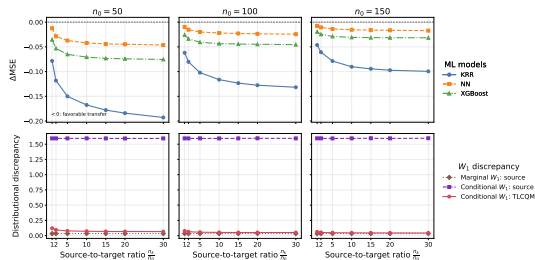
where $\theta = \left(1, \frac{1}{2}, \dots, \frac{1}{5} \right)^T \in \mathbb{R}^5$, $X^{(1)}, X^{(2)} \sim \mathcal{N}(\mathbf{1}_5, I_5)$, and $\epsilon \sim \mathcal{N}(0, 0.25)$.

- Target Domain: $Y^{(0)} = \frac{1}{3} \sin \left(3\theta^T X^{(0)} \right) - 3 + \epsilon$ with $X^{(0)} \sim \mathcal{N}(\mathbf{0}_5, 0.25 \cdot I_5)$ and $\epsilon \sim \mathcal{N}(0, 0.25)$.

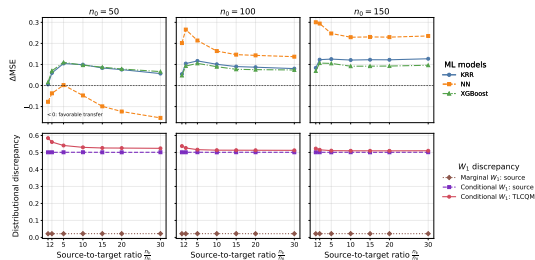


Diagnostics: When Does Transfer Help?

Reversed conditional relation



Target outside the affine span



Conditional alignment, not marginal similarity, determines transfer:
TLCQM corrects a reversible conditional mismatch, whereas persistent bias outside the calibrated family can produce negative transfer.